0.5 Project Scope Sheet · Group 1 · Fernão de Oliveira

11 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 1 · Fernão de Oliveira
Members (student number)	106238 · 107304 · 106645
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP01 · Longitudinal Strength Calculation
Specialization 2	SP02 · Propulsion Matching
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group1_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

How the work is split. The in-class exercises are individual practice of each method on the reference ship (the 37,600 t oil/chemical tanker) and are not graded. This project is your group's work, done outside class: you apply each method to the Azores Feeder and complete the full scope of every deliverable below. A report that only repeats the class exercises does not meet the scope.

Planning (advised). A work plan is not a deliverable, but plan internally: a rough approach and a split of the tasks across the group, so that the work, the presentation and the knowledge of the whole project are shared evenly by every member. You are welcome to share the plan and discuss it with the instructor.

Code flowchart (mandatory). Include a flowchart of your code that shows how the information flows: inputs, each calculation step, the iterations and the outputs. Use the symbols of ISO 5807, the standard for data, program and system flowcharts.

Common deliverables D1–D10

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)
D1	Design Basis	Ex. 1.2 does the owner's brief add up, classify the ship · Ex. 1.3 design phases and documents	130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked
D2	Reference Ship Database	Ex. 1.4 first Δ, L, B, T, D from the App. A regressions, with 95% bands	At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds
D3	Ship Model	Ex. 1.5 displacement equation · Ex. 2.3 main dimensions · Ex. 2.4 power · Ex. 3.1 weights · Ex. 3.2 capacity · Ex. 3.3 engine and propeller	Volume-carrier sequence · inputs, outputs and calculation sequence · open-water propeller efficiency η_0 as a stated, justified assumption (0.60) · electric drive, two shafts · batteries as deadweight · crane in the lightship · closed weight balance
D4	Voyage and Financial Model	Ex. 3.4 round voyage and fuel · Ex. 4.1 building cost · Ex. 4.2 RFR and NPV	Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)
D5	Validation Note	No class exercise: done by the group	Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed
D6	Optimization Report	Ex. 5.1 synthesis model · Ex. 5.2 optimization (demonstration + individual step) · Ex. 5.3 multi-objective trade-off	Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements
D7	Compliance Sheet	Ex. 4.3 GM, freeboard and attained EEDI	EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules (BV) · stability while the crane lifts · freeboard · draught at every port
D8	General Arrangement and Hull Form	Ex. 5.4 sectional area curve and LCB · Ex. 6.1 sections, bow and stern · Ex. 6.2 GA and capacity plan	Deck container layout and crane reach · battery slots aft · hull form of your choice, justified (a PSV-type hull is only a suggestion) · A3 sketch (profile and midship section) · hydrostatics check
D9	Specialization Documents (2)	No class exercise: self-study decks	One calculation note per specialization, with its required drawings (see Your specializations)
D10	Datasheet and Presentation	No class exercise	One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes

Exercises 7.1 and 7.2 (digital transition and AI) come after the deadline: your group may show them in the presentation, but they are not part of the ZIP.

Your studies and drawings

Your group's list from the Statement of Work: 32 items, 5 of them drawings. Tick each one off as it is done.

ID	Title	Deliverable	Status
STU-D1.1	Owner's requirements and port limits	D1	
STU-D1.2	Environmental targets	D1	
STU-D2.1	Reference ship database with sources	D2	
STU-D2.2	Regressions and bounds	D2	
STU-D3.1	Ship model	D3	
STU-D3.2	Closed weight balance	D3	
STU-D4.1	Voyage model	D4	
STU-D4.2	Financial model	D4	
STU-D5.1	Validation against reference ships	D5	
STU-D6.1	Optimization	D6	
STU-D7.1	EEDI and CII	D7	
STU-D7.2	Freeboard and intact stability	D7	
STU-D7.3	Draught at every port	D7	
STU-D8.1	Hydrostatics check	D8	
DWG-D8.1	Profile (A3)	D8	
DWG-D8.2	Midship section (A3)	D8	
STU-D10.1	Datasheet (template 0.4)	D10	

ID	Title	Deliverable	Status
STU-D10.2	Presentation, 15 minutes	D10	
STU-B.1	Capacity count on own arrangement	D8	
DWG-B.1	Container and battery layout with crane reach	D8	
STU-SP01.1	Lightship and deadweight distribution	D9 · SP01	
STU-SP01.2	Four critical loading conditions	D9 · SP01	
STU-SP01.3	Still-water and wave bending moments and shear forces	D9 · SP01	
STU-SP01.4	Allowable stresses, bending and shear	D9 · SP01	
STU-SP01.5	Preliminary scantlings of the midship cross-section	D9 · SP01	
STU-SP01.6	Section modulus, moment of inertia and neutral axis	D9 · SP01	
STU-SP01.7	Allowable still-water bending moment and shear force vs loading conditions	D9 · SP01	
DWG-SP01.1	Midship cross-section	D9 · SP01	
STU-SP02.1	Optimum B-series propeller	D9 · SP02	
STU-SP02.2	Engine matching	D9 · SP02	
STU-SP02.3	PTO/PTI or electric operating modes	D9 · SP02	
DWG-SP02.1	Stern arrangement	D9 · SP02	

Your specializations

One calculation note per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

- STU-SP01.1 Lightship weight and deadweight distribution along the length.
- STU-SP01.2 Four critical loading conditions (subject to justification and/or definition with the instructor).
- STU-SP01.3 Still-water and wave bending moments and shear forces for each loading condition.
- STU-SP01.4 Allowable stresses, in bending and in shear.
- STU-SP01.5 Preliminary scantlings of the midship cross-section.
- STU-SP01.6 Section modulus, moment of inertia and neutral axis of the midship section.
- STU-SP01.7 Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them.
- DWG-SP01.1 Midship cross-section.

SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

- STU-SP02.1 Optimum Wageningen B-series propeller for the service speed.
- STU-SP02.2 Engine matching: operating points on the motor layout diagram, with the sea margin.
- STU-SP02.3 The two-shaft electric modes (Baseline B).
- DWG-SP02.1 Stern arrangement: propeller, shaft line and rudder.

The full text of every item is in the Statement of Work.